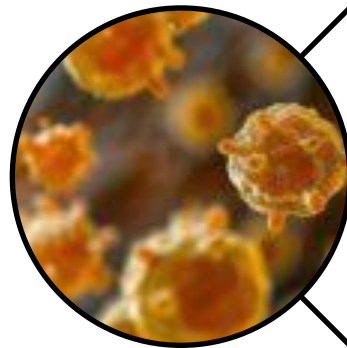


connecting people with science

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KEMRI/WRAIR-Africa, Kenya
CP3 – Quality Control of Sequence Reads

March 2026



**Antimicrobial
Resistance of Bacterial
Pathogens – Africa**

7-13 March 2026

NICD, Johannesburg,
South Africa

Outline

- Introduction
- Short Read QC (FastQC)
- Long Read QC (NanoQC)
- Preprocessing raw reads
 - Trimming
 - Filtering
 - Deduplication
 - Contamination Removal
- Post trim/filter QC

Data Retrieval from Public databases

Obtain sequencing reads from: Public repositories (ENA, SRA, NCBI)

Typical formats: (FASTQ or FASTQ.gz)

Example tools/approaches:

- wget
- SRA-Toolkit
 - fasterq-dump
 - prefetch

INTRODUCTION



- ❑ Errors at any of these steps can negatively impact the quality of the sequence
- ❑ If these errors are not removed from the raw reads, they might be incorporated into your analysis output and would be harder to resolve later on.
- ❑ Therefore, it is important to perform quality checks on the raw sequence reads before starting your analysis.

Quality Control and Preprocessing

- **Key idea:**

QC transforms raw sequencing data → high-quality reads ready for analysis.

What Do We Remove During Read Preprocessing?

- **Adapter sequences**
 - Artificial sequences added during library preparation
 - Appear when DNA fragments are shorter than the sequencing read length
- **Barcodes / index sequences**
 - Short sequences used to identify samples in multiplexed sequencing
 - Removed to retain only the biological DNA sequence

What Do We Remove During Read Preprocessing?

- **Low-quality bases**
 - Bases with low confidence scores (often at the ends of reads)
 - Trimming improves overall read quality
- **Very short reads**
 - Reads that become too short after trimming
 - Often unreliable for mapping or assembly
- **Contaminant reads**
 - DNA from unintended sources (e.g., host DNA, environmental organisms, PhiX control)
 - Removed to avoid bias in downstream analysis

Tools for Read Quality Control and Preprocessing

1. **FastQC**

- Assess quality of Illumina sequencing reads
- Provides metrics such as base quality, GC content, and sequence duplication

2. **NanoQC / NanoPlot**

- Quality assessment for Oxford Nanopore sequencing reads
- Summarizes read length, quality score distribution, and yield



wdecoster/
NanoPlot

Tools contd..

3. **fastp**

- Performs adapter trimming, quality trimming, and filtering
- Generates both processed reads and QC reports

4. **Porechop**

- Removes adapter sequences from Nanopore reads
- Detects and splits chimeric reads

5. **Filtlong**

- Filters long reads based on length and quality
- Retains the highest-quality reads for downstream analysis

6. **MultiQC**

- Aggregates results from multiple QC tools into a single summary report

The logo for MultiQC, featuring the text "MultiQC" in a bold, black, sans-serif font. The "Q" is stylized with a white plus sign inside it. Below the text are three horizontal lines in red, green, and blue.The logo for Filtlong, featuring the text "Filtlong" in a black, sans-serif font. Below the text is a stylized illustration of a hot dog in a bun with red and yellow mustard.The logo for Porechop, featuring the text "Porechop" in a black, sans-serif font. Below the text is a stylized illustration of a DNA double helix being cut by a knife.

FastQC

- ❑ FastQC: A tool for quality control of high-throughput sequence data.
- ❑ The tool can be run by both command-line and also has a graphical user interface
- ❑ The tool provides you with a report on the quality of sequence reads using a traffic light system, red, amber and green.
- ❑ There are a number of parameters which we will learn in this module that help us in assessing the sequence data quality.

FastQC

Summary

- ✓ [Basic Statistics](#)
- ✗ [Per base sequence quality](#)
- ! [Per tile sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ✗ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ✓ [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ! [Overrepresented sequences](#)
- ✓ [Adapter Content](#)
- ! [Kmer Content](#)

- **Green:** Indicates no significant issues detected.
- **Amber:** Suggests potential problems that may require attention.
- **Red:** Indicates critical issues that need immediate investigation and possibly corrective action.

Basic Statistics



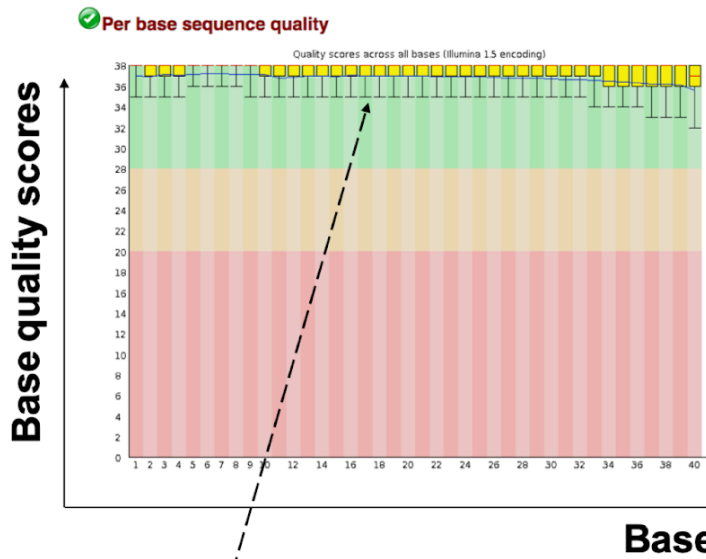
Basic Statistics

Measure	Value
Filename	ERR2093245_1.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1339517
Sequences flagged as poor quality	0
Sequence length	250
%GC	51

- Contain basic information from sequence reads like number of reads, length, GC%

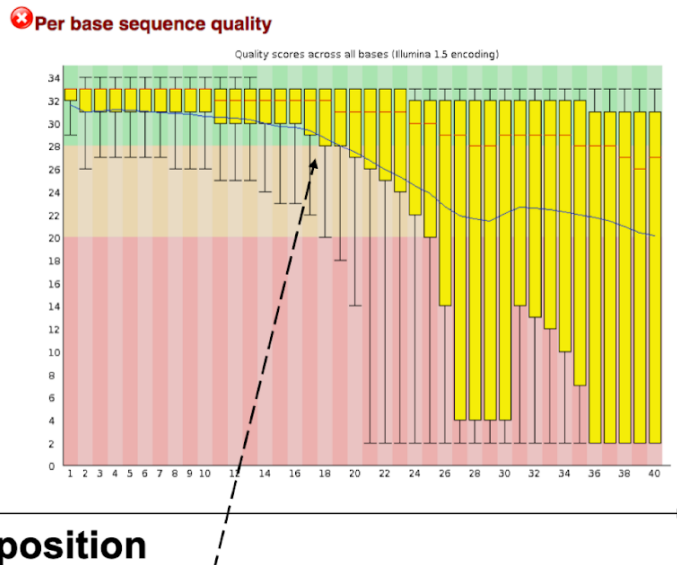
Quality

a. Good quality



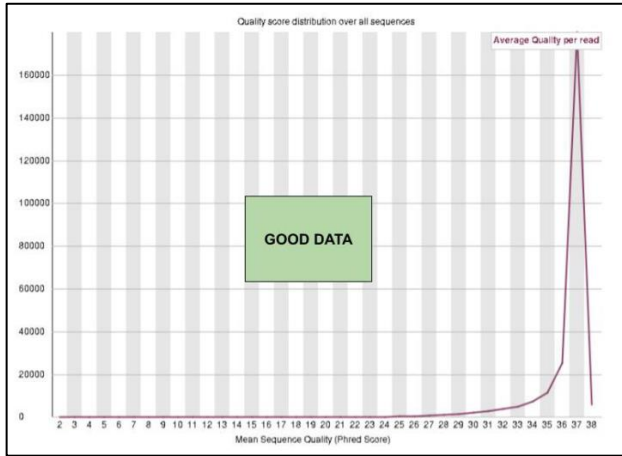
The quality scores for all the bases are in green zone

b. Bad quality

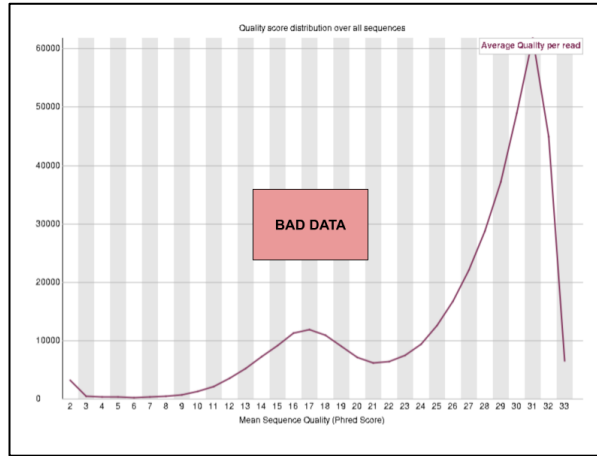


Less than half of the bases have quality scores in the green zone

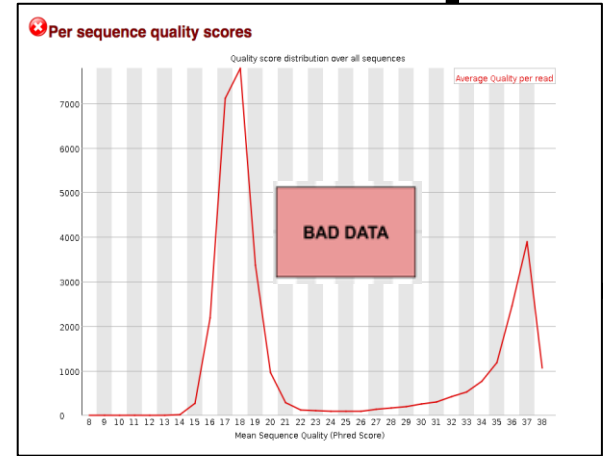
Per sequence quality scores



Single peak at av.
quality score > 27



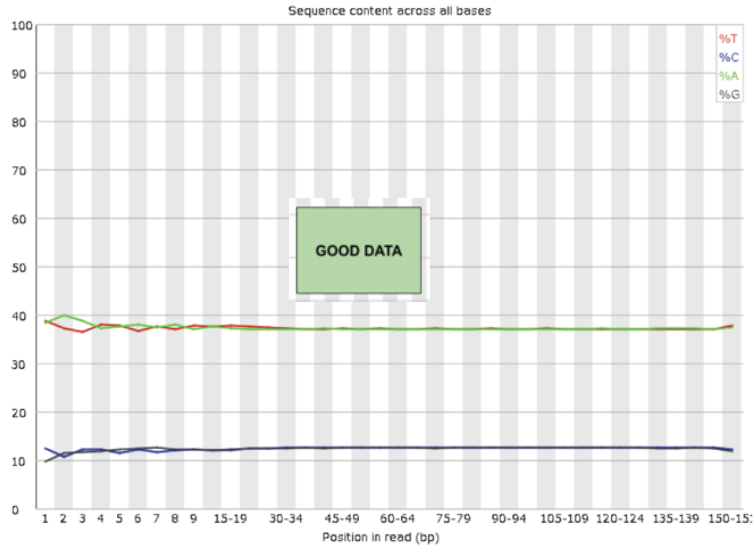
Bimodal distribution –
Contaminating artifacts?



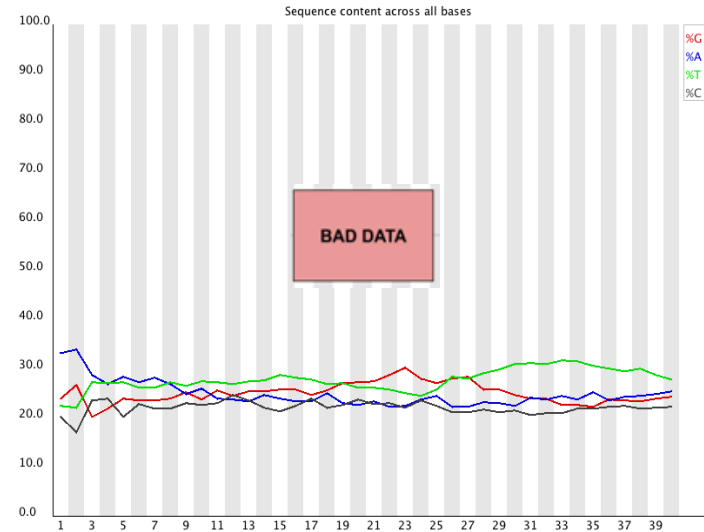
Majority of reads have
low quality

Per base sequence content

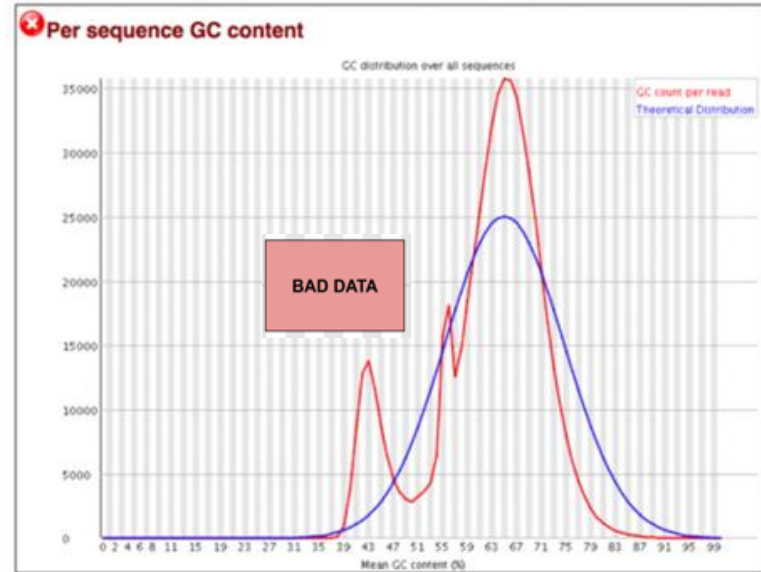
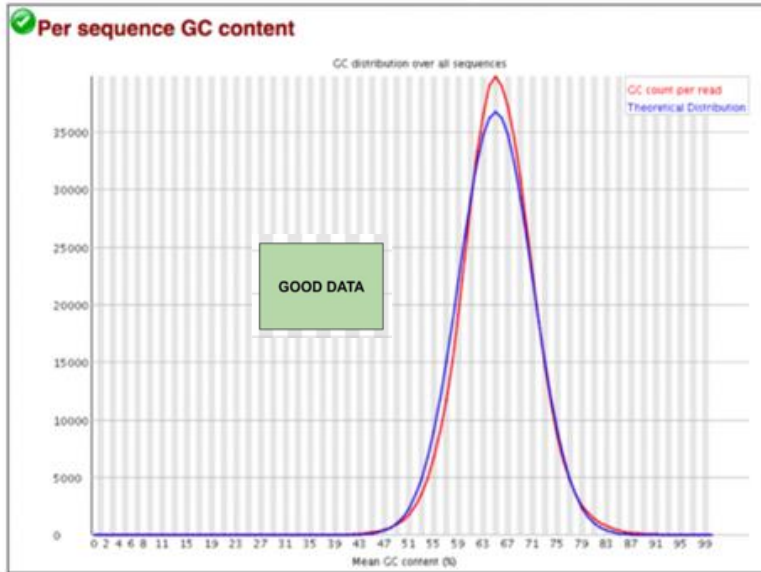
✓ Per base sequence content



✗ Per base sequence content

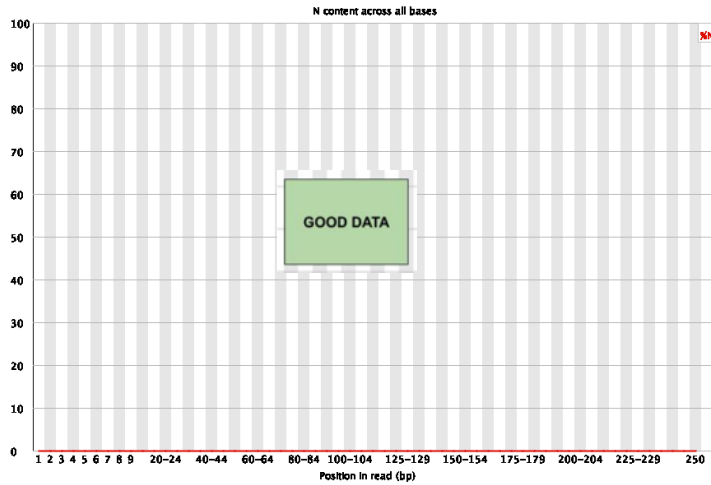


Per sequence GC content



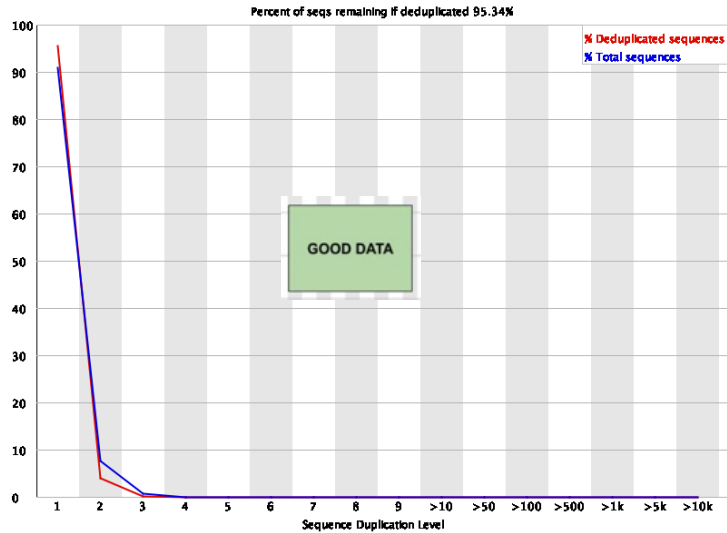
Per base N content

✓ Per base N content

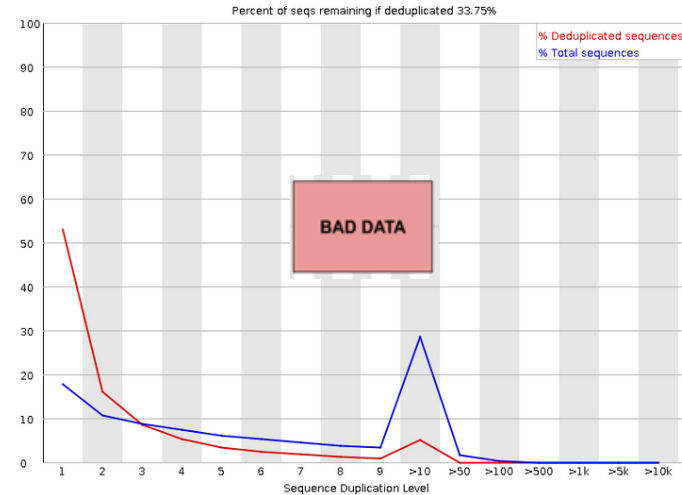


Sequence duplication levels

✔ Sequence Duplication Levels

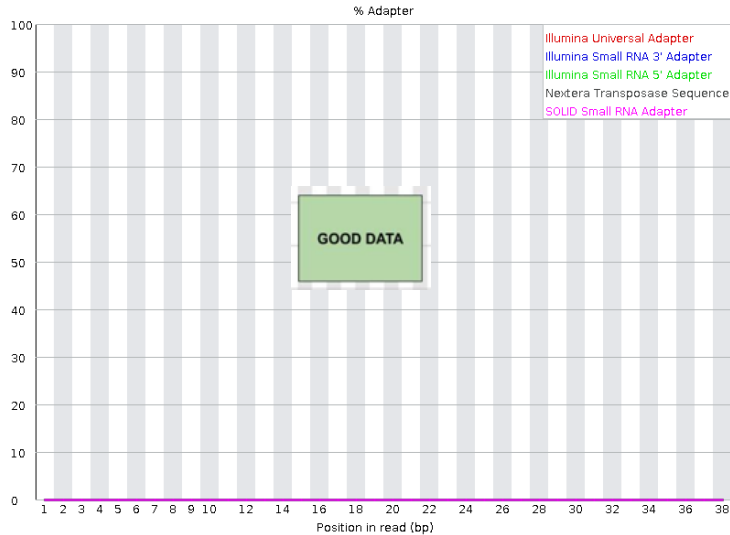


✖ Sequence Duplication Levels

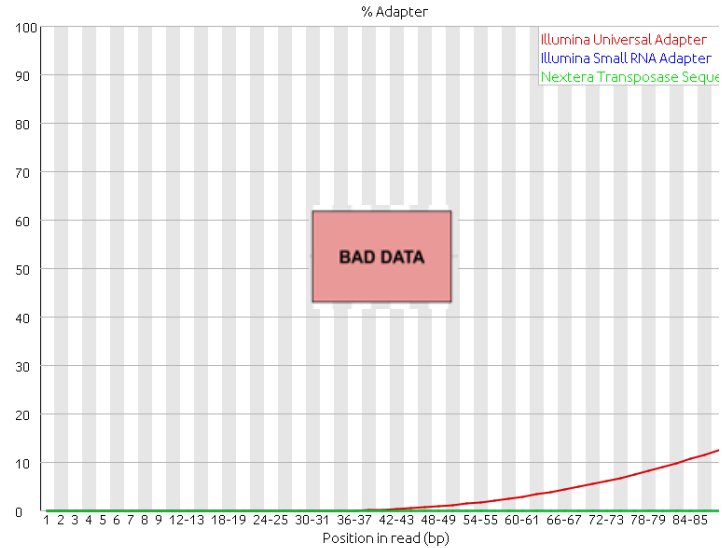


Adapter content

✓ Adapter Content



✗ Adapter Content



NanoQC

- NanoQC is a lightweight tool for quality assessment of Oxford Nanopore sequencing data.
- It analyzes FASTQ files and generates summary statistics describing read quality and length.
- Helps evaluate sequencing performance before preprocessing.

Long-read Trimming (Porechop)

Porechop is a tool used to remove adapter sequences from Oxford Nanopore reads.

Detects and removes Nanopore adapter sequences.

Identifies and splits chimeric reads (reads containing two fragments joined together).

Prevents adapters from interfering with:

- genome assembly
- read mapping
- variant detection

Long-read Filtering (Filtlong)

Filtlong is a tool used to filter Oxford Nanopore long reads based on quality and length.

Nanopore sequencing produces reads with variable quality and length, so filtering helps retain the most useful reads for analysis.

Removes low-quality reads.

Filters out very short reads that may not be useful for assembly or mapping.

Selects the best subset of reads based on quality and length.

Deduplication (Dedupe.sh)

Deduplication is the process of identifying and removing duplicate sequencing reads from a dataset.

Duplicate reads often arise during PCR amplification during library preparation.

Why Deduplication is Important

Prevents **overestimation of read coverage**.

Reduces bias in downstream analyses such as:

- variant calling
- abundance estimation
- genome assembly

Contaminant Removal

1. Common Sources of Contamination
2. Host DNA (e.g., human DNA in clinical samples)
3. Environmental microorganisms
4. PhiX control sequences used in Illumina sequencing
5. DNA from other samples due to cross-contamination

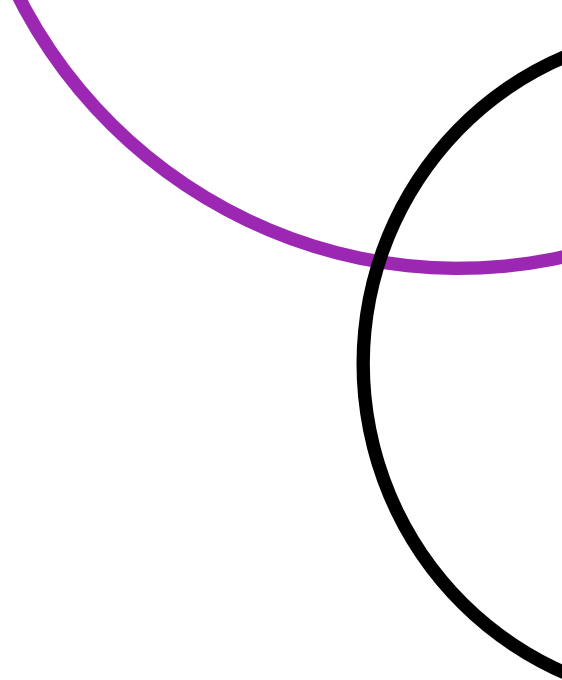
Why Contamination Removal is Important

1. Prevents incorrect species identification
2. Improves accuracy of genome assembly and mapping
3. Avoids false detection of genes, including AMR genes

Post-trimming/filtering QC

After trimming and filtering, sequencing reads should be re-evaluated to confirm improvements in data quality.

Post-trimming QC ensures that preprocessing steps successfully removed unwanted sequences and low-quality reads.



Acknowledgements and Partners



thank you!

Please contact wellcomeconnectingscience.org
for more information.

